

Ellipses



Standard form

- 1 For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find:
 - a the vertices
 - b the foci
 - c the endpoints of the minor axis
 - d the eccentricity
 - 2 Write the standard-form equation of the ellipse with foci $(0, \pm 3)$ and vertices $(0, \pm 5)$.
 - 3 Consider the ellipse $9x^2 + 4y^2 = 36$.
 - a Write the equation in standard form.
 - b State the lengths of the major and minor axes.
 - c Find the coordinates of the foci.
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Applications

- 4 A whispering gallery has an elliptical ceiling 40 m long and 24 m high at the center (so $a = 20$ and $b = 12$, with the major axis horizontal). Two people stand at the foci. How far apart are they, to one decimal place?
- 5 An ellipse centered at the origin has a horizontal major axis of length 12 and passes through the point $(3, 2\sqrt{3})$. Find its equation.